

2012

6 4 24

1 4

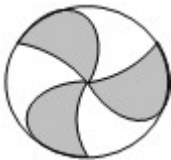
- A $a^4 \cdot a^3 = a^7$ B $a^6 \div a^3 = a^3$
C $a^3 \cdot a^2 = a^5$ D $a^3 \cdot b^3 = a \cdot b^3$

2 4 2 4

- A $y = 2x$ B $y = \frac{1}{2x}$ C $y = \frac{8}{x}$ D $y = x^2$

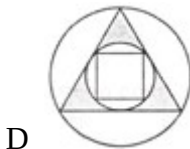
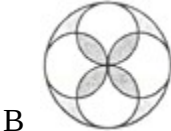
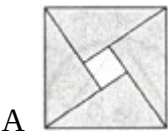
3 4

6



- A $\frac{1}{6}$ B $\frac{1}{3}$ C $\frac{1}{2}$ D $\frac{2}{3}$

4 4



- A B C D

5 4 ABCD $\angle A = \angle B = \angle C = 90^\circ$

- A $\angle D = 90^\circ$ B $AB \parallel CD$ C $AD \parallel BC$ D $BC \perp CD$

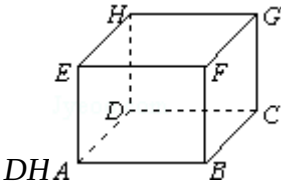
6 4

- A 10% 100

- B ABCD EFGH EF FG

- C 540°

D



12 4 48

7 4 2^2 _____

8 4 $\sqrt{x^2-1}=2$ _____

9 4 $\frac{x^2+1}{x} - \frac{2x}{x^2+1} = 3$ $\frac{x}{x^2+1} = y$ y _____

10 4 $x^2 - 2x + a = 0$ a _____

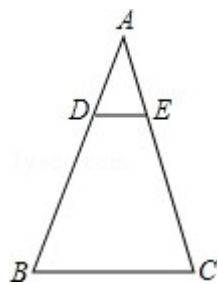
11 4 $y = kx + 1$ $x = k$ _____

12 4 $a = x$ _____

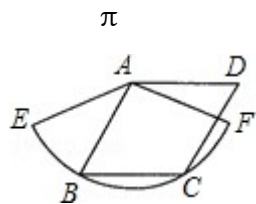
13 4 $\triangle ABC$ G BC D 2 AD _____

14 4 5 7 _____

15 4 $\triangle ABC$ $DE \parallel BC$ $DE = 1$ $BC = 4$ $\triangle ADE \sim \triangle ABC$ _____

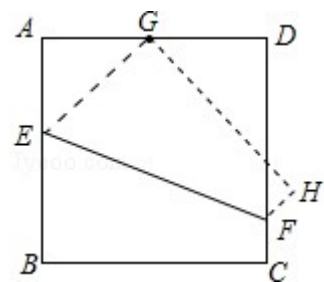


16 4 1 $ABCD$ $B = C$ AEF EF BC _____



17 4 $ABCD$ $|\vec{AB}| = 2$ $|\vec{BC}| = 1$ $|\vec{AB} + \vec{BC}|$ _____

18 4 4 $ABCD$ EF B AD G $BCFE$

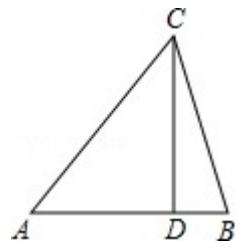


□□□□□□□□□□ 7 □□□□ 78 □□

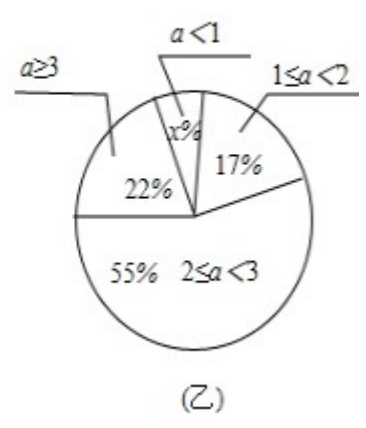
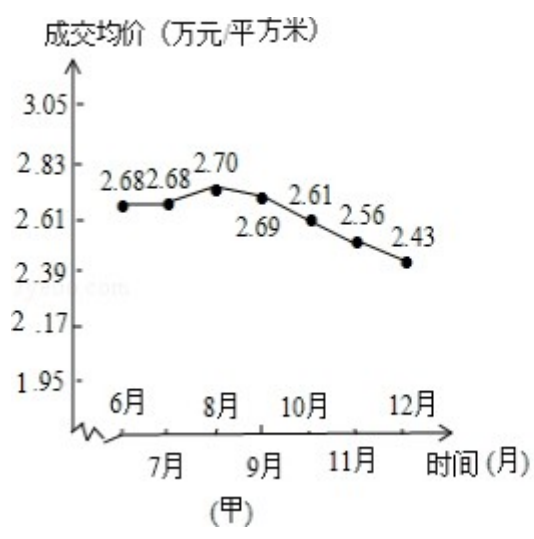
19 10 $(\frac{a^2-2a+1}{a^2-1}+\frac{1}{a})\div\frac{1}{a+1}$ $a=\sqrt{2}$

20 10 $\begin{cases} x^2+5xy+6y^2=0 \\ x+y=2 \end{cases}$

21 10 $\triangle ABC$ $CD\perp AB$ $\sin A=\frac{4}{5}$ $AB=13$ $CD=12$ AD $\tan B$



22 10 2011 6 12 /
2011 12 a /



1 2011 6 2011 12

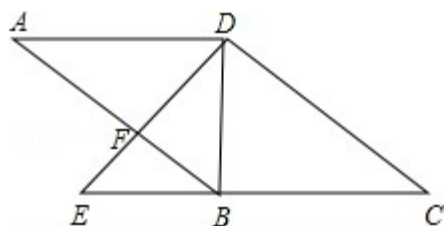
2 x _____

3 2011 12 2400 200 12 60000 2

23 12 $ABCD$ $AD\parallel BC$ E CB DE AB F DB $\angle AFD$
 $\angle DBE$ $DE^2=BE\cdot CE$

1 $\angle DBE=\angle CDE$

2 $BD \perp \angle ABC$ $ABCD$

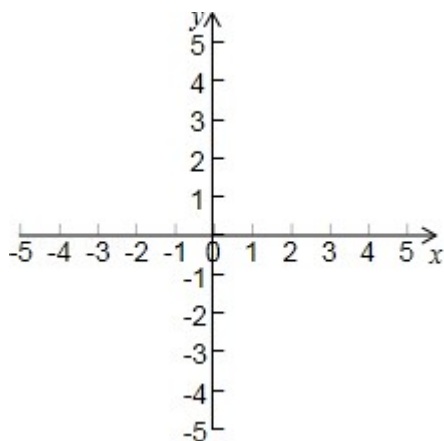


24 12 $y = \frac{1}{6}x + 2\sqrt{3}$ A y B AB ABC

1 AB C

2 M m 1 $\triangle ABM$ $\triangle ABC$ M

3 x N 1 C CM N



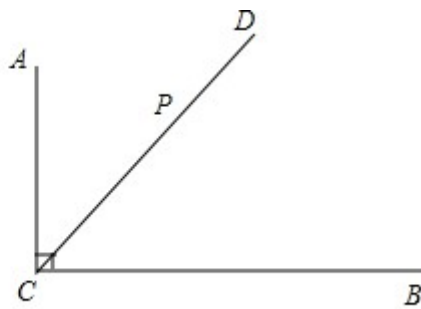
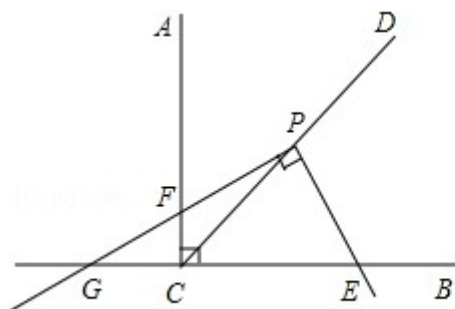
25 14 $\angle ACB = 90^\circ$ $CD \perp \angle ACB$ P CD $CP = \sqrt{2}$ P
 P CB E CA CB F G

1 F CA

① $PF \perp PE$

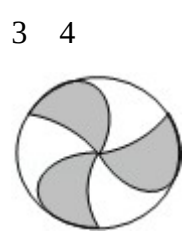
② $CF \perp EG$ y y x

2 $EF \perp \triangle CEF \triangle EGP$ EG



备用图

A D



6

A $\frac{1}{6}$

B $\frac{1}{3}$

C $\frac{1}{2}$

D $\frac{2}{3}$

X5

$\frac{1}{2}$

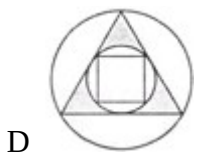
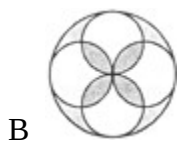
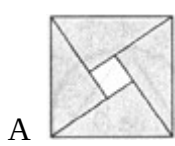
$\frac{1}{2}$

C

A

A

4 4



P3

R5

A

B

C

D

A

180

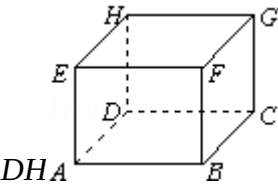
5 4 $ABCD$ $\angle A$ $\angle B$ $\angle C$ 90°
A $\angle D$ 90° B AB CD C AD BC D BC CD
LF
16

$\angle A$ $\angle B$ $\angle C$ 90° D
①

②

6 4
A 10% 100

B $ABCD$ $EFGH$ EF FG



C 540°

D
I1 L3 M2 X3
16 2B

A 10% 100
B $ABCD$ $EFGH$ EF FG DH
C 540°
D
B

□□□□□□□□□□ 12 □□□□ 4 □□□□ 48 □□

7 4 2^2 4
1E

2^2 2×2 4

$$1 \quad 1 \quad 1$$

$$8 \quad 4 \quad \sqrt{x^2-1}=2 \quad \text{--- } x=\pm\sqrt{5}\text{---}$$

AG

x

$$x^2-1=4$$

$$x^2=5$$

$$x=\pm\sqrt{5}$$

$$x=\pm$$

$$x=\pm\sqrt{5}$$

$$9 \quad 4 \quad \frac{x^2+1}{x}-\frac{2x}{x^2+1}=3 \quad \frac{x}{x^2+1}=y \quad y \quad \frac{2y^2+3y-1}{}$$

$$\frac{0}{}$$

B4

$$1$$

$$\frac{x}{x^2+1}=y$$

$$\therefore y=\frac{x}{x^2+1}$$

$$\therefore \frac{1}{y}=2y+3$$

$$2y^2+3y-1=0$$

$$2y^2+3y-1=0$$

$$10 \quad 4 \quad x^2-2x+a=0 \quad a=\frac{1}{}$$

AA

$$0 \quad a \quad a$$

$$\therefore x^2-2x+a=0$$

$$\therefore \Delta=4-4a \geq 0 \quad a \leq 1$$

$\triangle$

$$1 \triangle 0 \Leftrightarrow$$

$$2 \triangle 0 \Leftrightarrow$$

$$3 \triangle 0 \Leftrightarrow$$

$$11 \quad 4 \quad y \quad k \quad 1 \quad x \quad y \quad x \quad k \quad \underline{k \quad 1}$$

F7

33

$$k \quad k \quad 1 \quad 0$$

$$\therefore y \quad k \quad 1 \quad x \quad y \quad x$$

$$\therefore k \quad 1 \quad 0$$

$$k \quad 1$$

$$k \quad 1$$

$$k$$

$$y \quad kx$$

$$k$$

$$k \quad 0$$

$$y \quad x$$

$$k \quad 0$$

$$y \quad x$$

$$12 \quad 4 \quad a \quad x \quad \underline{a \quad 1 \quad x^2}$$

AC

123

$$\times 1$$

$$\times 1$$

$$a \quad 1 \quad x$$

$$a \quad 1 \quad x \quad 1 \quad x \quad a \quad 1 \quad x^2$$

$$a \quad 1 \quad x^2$$

$$a$$

$$x$$

$$a \quad 1 \pm x^2$$

$$13 \quad 4 \quad \triangle ABC \quad G \quad BC \quad D \quad 2 \quad AD \quad \underline{6}$$

K5

$$\therefore G \triangle ABC$$

$$\therefore AG \quad 2GD \quad 4$$

$$\therefore AD \quad AG + GD \quad 6$$

2

$$14 \quad 4 \quad 5 \quad 7 \quad \underline{9}$$

LL

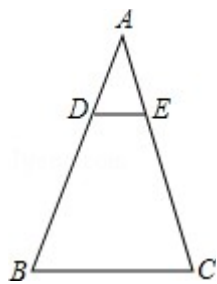
$$x \quad 5+x \quad 2 \times 7$$

$$x \quad 9$$

$$9$$

$$9$$

$$15 \quad 4 \quad \triangle ABC \quad DE \parallel BC \quad DE = \frac{1}{4} BC \quad \triangle ADE \sim \triangle ABC \quad \underline{\frac{1}{16}}$$



S9

$$DE \parallel BC$$

$$\because DE \parallel BC$$

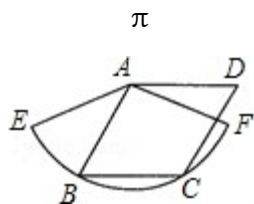
$$\therefore \triangle ADE \sim \triangle ABC$$

$$\therefore DE = \frac{1}{4} BC$$

$$\therefore \triangle ADE \sim \triangle ABC \quad \frac{1}{16}$$

$$\frac{1}{16}$$

$$16 \quad 4 \quad 1 \quad ABCD \quad B \quad C \quad AEF \quad EF \quad BC \quad -\frac{\pi}{3}$$



L8

MN

$$B \quad C$$

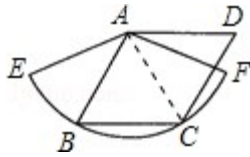
$$AEF \quad \widehat{EF}$$

$$B \quad C$$

$$AC \quad \triangle ABC$$

$$\widehat{BC}$$

$$AC$$



$$\because ABCD \quad AB \quad BC$$

$$\because AC \quad AB$$

$$\because AB \quad BC \quad AC \quad \triangle ABC$$

$$\because \angle BAC \quad 60^\circ$$

$$\because \widehat{BC} = \frac{60\pi \times 1}{180} = \frac{\pi}{3}$$

$$17 \quad 4 \quad ABCD \quad \begin{matrix} B \quad C \\ \overrightarrow{AB} \end{matrix} = 2 \quad \begin{matrix} AEF \quad \widehat{EF} \\ \overrightarrow{BC} \end{matrix} = 1 \quad \begin{matrix} B \quad C \\ \overrightarrow{AB} + \overrightarrow{BC} \end{matrix} = \sqrt{5}$$

LM *

$$\begin{matrix} ABCD \\ \overrightarrow{AB} + \overrightarrow{BC} \end{matrix} = \overrightarrow{AC} \quad \begin{matrix} \overrightarrow{AB} \\ \overrightarrow{BC} \end{matrix} = 2 \quad \begin{matrix} \overrightarrow{BC} \\ \overrightarrow{AC} \end{matrix} = 1 \quad \begin{matrix} AB \quad BC \\ \angle B \end{matrix} = 90^\circ \quad AC$$

$$\because \quad ABCD$$

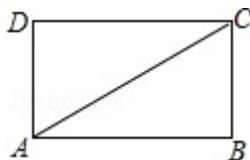
$$\because \angle B \quad 90^\circ$$

$$\because |\overrightarrow{AB}| = 2 \quad |\overrightarrow{BC}| = 1$$

$$\because AB = 2 \quad BC = 1$$

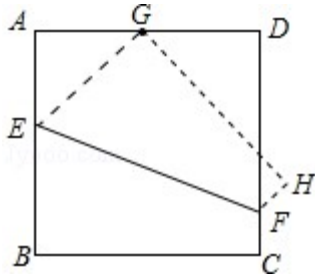
$$\because AC = \sqrt{AB^2 + BC^2} = \sqrt{5}$$

$$\because |\overrightarrow{AB} + \overrightarrow{BC}| = |\overrightarrow{AC}| = \sqrt{5}$$



$$18 \quad 4 \quad 4 \quad ABCD \quad EF \quad B \quad AD \quad G \quad BCFE$$

$$\underline{6}$$



PB

16

$$\frac{CF}{AE} = \frac{HF}{BE} = \frac{GE}{AE} = \frac{BE}{AE} = x$$

BE

AE

CF

C H

B G EF

$$\therefore \frac{CF}{AE} = \frac{HF}{BE} = \frac{GE}{AE}$$

$$\frac{BE}{AE} = x \quad AE = 4 \quad x$$

$$\because ABCD$$

$$\therefore \angle A = 90^\circ$$

$$\therefore AE^2 + AG^2 = EG^2$$

$$\because B \quad AD \quad G$$

$$\therefore AG = 2$$

$$\therefore 4x^2 + 2^2 = x^2$$

$$x = 2.5$$

$$\therefore BE = 2.5$$

$$\because ABCD$$

$$\therefore AB \parallel CD \quad \angle B = 90^\circ$$

$$\because E \quad F \quad AB \quad CD$$

$$\therefore BCFE$$

$$\because BE = GE = 2.5 \quad AB = 4$$

$$\therefore AE = 1.5$$

$$\therefore \sin \angle 1 = \frac{3}{5} \quad \tan \angle 1 = \frac{3}{4}$$

$$\because \angle 1 + \angle 2 = 90^\circ \quad \angle 2 + \angle 3 = 90^\circ$$

$$\therefore \angle 3 = \angle 1$$

$$\therefore \sin \angle 3 = \sin \angle 1 = \frac{3}{5}$$

$$\text{Rt}\triangle DGP \because \angle D = 90^\circ$$

$$DG = 2 \sin \angle 3 = \frac{DG}{GP} = \frac{3}{5}$$

$$\therefore PG = \frac{10}{3}$$

$$\therefore PH = GH = GP = \frac{2}{3}$$

$$\because \angle 4 = \angle 3$$

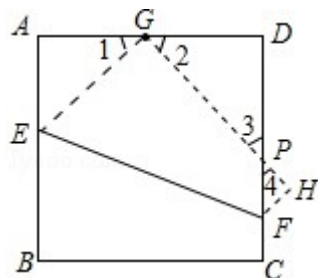
$$\therefore \tan \angle 4 = \tan \angle 3 = \tan \angle 1 = \frac{3}{4}$$

$$\text{Rt}\triangle HPF \because \angle H = \angle C = 90^\circ$$

$$\therefore FC = HF = \frac{1}{2}$$

$$\therefore S_{BCFE} = \frac{1}{2} (FC + BE) \times BC = \frac{1}{2} \times \left(\frac{1}{2} + 2.5 \right) \times 4 = 6$$

6



□□□□□□□□□□ 7 □□□□ 78 □□

$$19 \quad 10 \quad \left(\frac{a^2 - 2a + 1}{a^2 - 1} + \frac{1}{a} \right) \div \frac{1}{a+1} \quad a = \sqrt{2}$$

6D

11

a

$$\left(\frac{a-1}{a+1} + \frac{1}{a} \right) \cdot (a+1) = a-1 + \frac{a+1}{a} = \frac{a^2+1}{a}$$

13 22

$$a=\sqrt{2} \quad \frac{(\sqrt{2})^2+1}{\sqrt{2}} \quad \frac{3\sqrt{2}}{2}$$

$$20 \quad 10 \quad \begin{cases} x^2+5xy+6y^2=0 \\ x+y=2 \end{cases}$$

AF

11

①

$$\begin{cases} x^2+5xy+6y^2=0 \textcircled{1} \\ x+y=0 \textcircled{2} \end{cases}$$

$$\textcircled{1} \quad x+2y \quad x+3y \quad 0$$

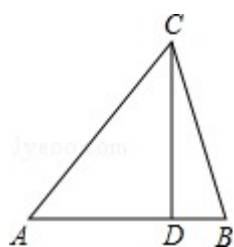
$$x+2y \quad 0 \quad x+3y \quad 0 \dots 4$$

$$\begin{cases} x+2y=0 \\ x+y=2 \end{cases} \quad \begin{cases} x+3y=0 \\ x+y=2. \end{cases}$$

$$\begin{cases} x_1=4 \\ y_1=-2 \end{cases} \quad \begin{cases} x_2=3 \\ y_2=-1. \end{cases}$$

①

$$21 \quad 10 \quad \triangle ABC \quad CD \perp AB \quad \sin A = \frac{4}{5} \quad AB = 13 \quad CD = 12 \quad AD \quad \tan B$$



T1

T7

$$\sin A = \frac{4}{5} \quad CD = 12$$

$$AC = 15$$

$$AD = 9 \quad BD = 4$$

$\tan B$

$$\because CD \perp AB$$

$$\therefore \angle CDA = 90^\circ \dots 1$$

$$\therefore \sin A \frac{CD}{AC} = \frac{4}{5}$$

$$\therefore AC = 15 \dots 2$$

$$\therefore AD = 9 \dots 3$$

$$\therefore BD = 4 \dots 4$$

$$\therefore \tan B \frac{CD}{BD} = 3 \dots 5$$

22

10

2011

12

2011

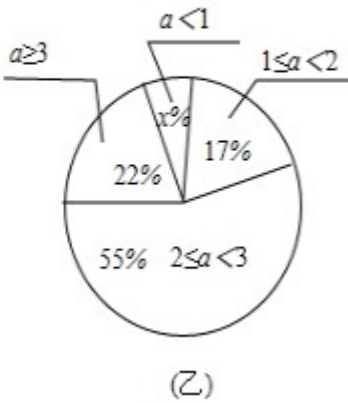
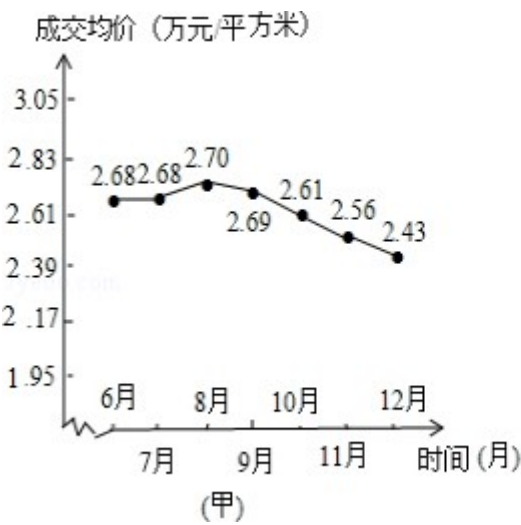
6

12

/

a

/



1

2011

6

2011

12

2

x

6

3

2011

12

2400

200

12

60000

2

VB

VD

W4

1

2011

6

2011

12

2

1

x

3

12

x

x

1

2011

6

2011

12

2.68

2.68

2.70

2.69

2.61

2.56

2.43

$$\therefore \frac{DE}{CE} = \frac{BE}{DE}$$

$$\because \angle E = \angle E$$

$$\therefore \triangle DBE \sim \triangle CDE$$

$$\therefore \angle DBE = \angle CDE$$

$$2 \because \angle DBE = \angle CDE$$

$$\because \angle DBE = \angle AFD$$

$$\therefore \angle CDE = \angle AFD$$

$$\therefore AB \parallel DC$$

$$\because AD \parallel BC$$

$$\therefore ABCD$$

$$\because AD \parallel BC$$

$$\therefore \angle ADB = \angle 1$$

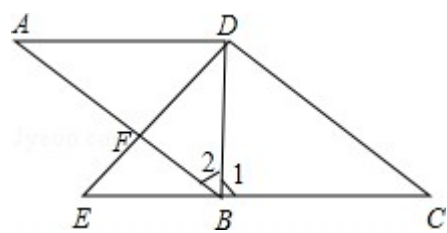
$$\because DB \perp AC$$

$$\therefore \angle 1 = \angle 2$$

$$\therefore \angle ADB = \angle 2$$

$$\therefore AB \perp AD$$

$$\therefore ABCD$$

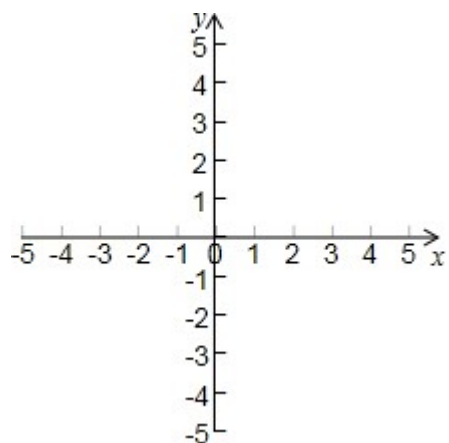


$$24 - 12 = y \cdot \frac{1}{6} (x + 2\sqrt{3})^2 \quad A = y \quad B = AB \quad ABC$$

$$1 \quad AB \quad C$$

$$2 \quad M = m \cdot 1 \quad \triangle ABM \quad \triangle ABC \quad M$$

$$3 \quad x \quad N \quad 1 \quad C \quad CM \quad N$$



HF

16 31 32

1 y C $\text{Rt}\triangle AOB$ 30°
 $\triangle ABC$ AC x B AC

C

2 “ M ” M “ $\triangle ABM$ ” $\triangle ABC$ “ AB ” C
 M AB $CM \parallel AB$ AB C

CM M

3 C CM C N ① CN ② CN

CN $\text{Rt}\triangle CAN$ AN N

1 $y = \frac{1}{6}x + 2\sqrt{3}$ $A(2\sqrt{3}, 0)$ y $B(0, 2)$

AB $y = kx + b$ $k \neq 0$

$k = \frac{\sqrt{3}}{3}$ $b = 2$ AB $y = \frac{\sqrt{3}}{3}x + 2$

$\angle BAO = 30^\circ \therefore \angle BAC = 60^\circ$

$\therefore \angle CAO = 90^\circ$

$\text{Rt}\triangle BAO$ $AB = 4$

$\therefore AC = 4$ $C(2\sqrt{3}, 4)$

2 $\therefore C, M$ $\triangle ABM$ $\triangle ABC$

$\therefore CM \parallel AB$

$$CM \quad y = \frac{\sqrt{3}}{3}x + m \quad C \quad (2\sqrt{3}, 4) \quad CM$$

$$m = 6$$

$$\therefore CM \quad y = \frac{\sqrt{3}}{3}x + 6$$

$$M \quad (5\sqrt{3}, 1)$$

$$3 \quad C \quad (2\sqrt{3}, 4) \quad M \quad (5\sqrt{3}, 1)$$

$$CM = \sqrt{(-2\sqrt{3} + 5\sqrt{3})^2 + (4 - 1)^2} = 6$$

$$\textcircled{1} \quad C \quad N \quad CN = CM + 1 = 7$$

$$\text{Rt}\triangle CAN \quad AN = \sqrt{CN^2 - CA^2} = \sqrt{7^2 - 4^2} = \sqrt{33}$$

$$\therefore ON = AN + OA = \sqrt{33} + 2\sqrt{3} \quad ON = AN + OA = \sqrt{33} + 2\sqrt{3}$$

$$N \quad (\sqrt{33} + 2\sqrt{3}, 0) \quad (\sqrt{33} + 2\sqrt{3}, 0)$$

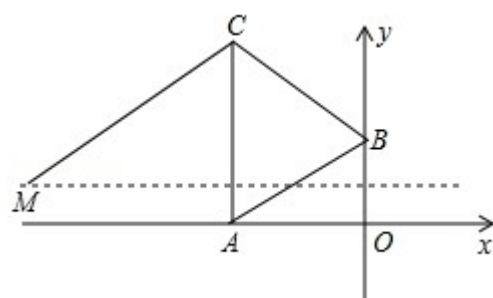
$$\textcircled{2} \quad C \quad N \quad CN = CM = 1 = 5$$

$$\text{Rt}\triangle CAN \quad CN = 5 \quad CA = 4 \quad AN = 3$$

$$\therefore ON = AN + OA = 3 + 2\sqrt{3} \quad ON = OA + AN = 2\sqrt{3} + 3$$

$$N \quad (3 + 2\sqrt{3}, 0) \quad (3 + 2\sqrt{3}, 0)$$

$$N \quad (3 + 2\sqrt{3}, 0) \quad (3 + 2\sqrt{3}, 0) \quad (\sqrt{33} + 2\sqrt{3}, 0) \quad (\sqrt{33} + 2\sqrt{3}, 0)$$



$$25 \quad 14 \quad \angle ACB = 90^\circ \quad CD \perp \angle ACB \quad P \quad CD \quad CP = \sqrt{2} \quad P$$

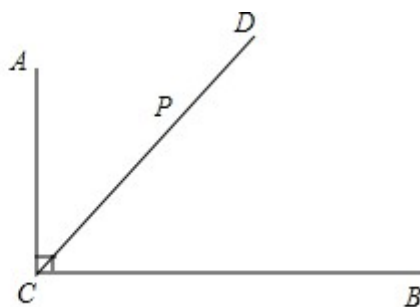
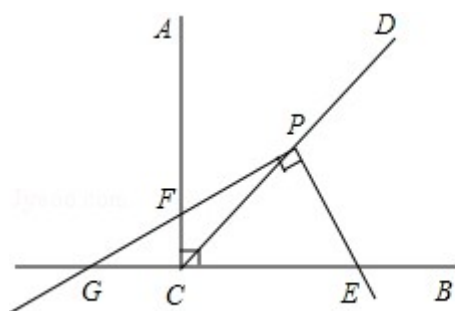
$$P \quad CB \quad E \quad CA \quad CB \quad F \quad G$$

$$1 \quad F \quad CA$$

$$\textcircled{1} \quad PF = PE$$

$$\textcircled{2} \quad CF \times EG = y \times x$$

$$2 \quad EF \quad \triangle CEF \quad \triangle EGP \quad EG$$



备用图

KD

KF

R2

S9

153

16

$$1 \quad \textcircled{1} \quad P \quad PM \perp AC \quad PN \perp BC \quad M \quad N \quad \triangle PMF \cong \triangle PNE \quad PF = PE$$

$$\textcircled{2} \quad \textcircled{1} \quad y \times x$$

$$2 \quad \triangle CEF \quad \triangle EGP \quad F \quad \textcircled{1} \quad F \quad CA \quad \textcircled{2} \quad F \quad AC$$

EG

1

$$\textcircled{1} \quad P \quad PM \perp AC \quad PN \perp BC \quad M \quad N$$

$$\because \angle CD = \angle ACB$$

$$\therefore PM = PN$$

$$\angle PMC = \angle MCN = \angle CNP = 90^\circ \quad \angle MPN = 90^\circ$$

$$\therefore \angle 1 + \angle FPN = 90^\circ$$

$$\because \angle 2 + \angle FPN = 90^\circ$$

$$\therefore \angle 1 = \angle 2$$

$$\therefore \triangle PMF \cong \triangle PNE$$

$$\therefore PF = PE$$

$\textcircled{2}$

$$\because CP = \sqrt{2}$$

$$\therefore CN = CM = 1$$

$$\because \triangle PMF \cong \triangle PNE$$

$$\therefore NE = MF = 1 \times x$$

$$\therefore CE = 2 - x$$

$$\therefore CF \parallel PN$$

$$\therefore \triangle GCF \sim \triangle GNP$$

$$\therefore \frac{CF}{PN} = \frac{CG}{GN}$$

$$\therefore CG = \frac{x}{1-x}$$

$$\therefore y = \frac{x}{1-x} + 2 - x \quad 0 \leq x \leq 1$$

$$2 \triangle CEF \triangle EGP \quad F$$

$$\textcircled{1} \quad F \quad CA$$

$$\therefore \angle GPE = \angle FCE = 90^\circ \quad \angle 1 \neq \angle PEG$$

$$\therefore \angle G < \angle 1$$

$$\therefore FG < FE$$

$$\therefore CG < CE$$

$$Rt\triangle EGP \quad EG = 2CP = 2\sqrt{2}$$

$$\textcircled{2} \quad F \quad AC$$

$$\therefore \angle GPE = \angle FCE = 90^\circ \quad \angle 1 \neq \angle 2$$

$$\therefore \angle 3 < \angle 2$$

$$\therefore \angle 1 = 45^\circ + \angle 5 < \angle 1 = 45^\circ + \angle 2$$

$$\therefore \angle 5 < \angle 2$$

$$\angle 3 < \angle 4 \quad \angle 5 < \angle 4$$

$$\therefore FC < CP = \sqrt{2}$$

$$\therefore FM = 1 + \sqrt{2}$$

$$\triangle PMF \cong \triangle PNE$$

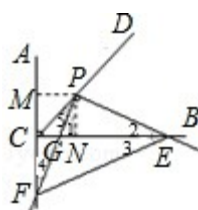
$$EN = 1 + \sqrt{2}$$

$$\therefore CF \parallel PN$$

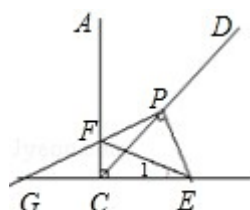
$$\therefore \frac{CF}{PN} = \frac{CG}{GN}$$

$$\therefore GN = \sqrt{2} + 1$$

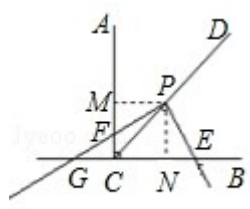
$$\therefore EG = 2\sqrt{2}$$



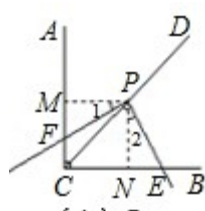
(2) ②



(2) ①



(1) ②



(1) ①